

Ludwig von Mises on Probability, Measurement, and Econometrics: Passages for a Note on Model Ambiguity

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Abstract

This note collects passages from Ludwig von Mises on probability, measurement, and econometrics. Mises distinguished class probability from case probability, denied that constant relations exist among economic magnitudes, and argued that statistical series are historical records rather than estimates of parameters. Section 2 collects what John Stuart Mill wrote about the calculation of chances, because Mises quotes Mill and because Mill's chapter says more than the quotation suggests. I assembled the passages for use in work with Lars Peter Hansen on model ambiguity. Quotations are verbatim from the editions listed in the bibliography. Section 7 states what Mises claimed and what he did not claim.

1 Class Probability and Case Probability

Chapter VI of *Human Action* divides probability in two [von Mises, 1998, pp. 105–118].

There are two entirely different instances of probability; we may call them class probability (or frequency probability) and case probability (or the specific understanding of the sciences of human action). The field for the application of the former is the field of the natural sciences, entirely ruled by causality; the field for the application of the latter is the field of the sciences of human action, entirely ruled by teleology. [von Mises, 1998, p. 107]

The two definitions follow.

Class probability means: We know or assume to know, with regard to the problem concerned, everything about the behavior of a whole class of events or phenomena; but about the actual singular events or phenomena we know nothing but that they are elements of this class. [von Mises, 1998, p. 107]

Case probability means: We know, with regard to a particular event, some of the factors which determine its outcome; but there are other determining factors about which we know nothing. [von Mises, 1998, p. 110]

Mises describes the calculus of probability as a translation of what the analyst already knows.

For this defective knowledge the calculus of probability provides a presentation in symbols of the mathematical terminology. It neither expands nor deepens nor complements our knowledge. It translates it into mathematical language. Its calculations repeat in algebraic formulas what we knew beforehand. [von Mises, 1998, p. 108]

He states the restriction on numerical evaluation flatly.

Case probability is not open to any kind of numerical evaluation. [von Mises, 1998, p. 113]

Section 5 of the chapter examines four statements a person might have made before the 1944 presidential election, among them “I estimate Roosevelt’s chances as 9 to 1.” Mises grants that the statement means something and denies that it is a probability:

It is a metaphorical expression. [von Mises, 1998, p. 114]

Section 4 closes with a passage that bears directly on decision making under ambiguity.

The necessity to adjust his actions to other people’s actions makes him a speculator for whom success and failure depend on his greater or lesser ability to understand the future. Every investment is a form of speculation. There is in the course of human events no stability and consequently no safety. [von Mises, 1998, p. 113]

The same section separates three ways of confronting the future. The gambler knows the frequency of outcomes in a class and nothing about his own case. The engineer

“knows only soluble problems and problems which cannot be solved under the present state of knowledge.” The actor who must anticipate other actors is neither [von Mises, 1998, p. 112].

Mises also quotes John Stuart Mill on the misuse of the calculus.

The history of every branch of knowledge records instances of the misapplication of the calculus of probability which, as John Stuart Mill observed, made it “the real opprobrium of mathematics.” [von Mises, 1998, pp. 106–107]

2 John Stuart Mill on the Calculus of Chances

Mises attributes the phrase “the real opprobrium of mathematics” to Mill. The phrase occurs in *A System of Logic*, Book III, Chapter XVIII, “Of the Calculation of Chances” [Mill, 1882]. Mill’s chapter says four things that bear on our subject.

2.1 What Mill wrote

Mill’s sentence assigns a cause, and the cause is the sentence before it.

It is obvious, too, that even when the probabilities are derived from observation and experiment, a very slight improvement in the data, by better observations, or by taking into fuller consideration the special circumstances of the case, is of more use than the most elaborate application of the calculus to probabilities founded on the data in their previous state of inferiority. The neglect of this obvious reflection has given rise to misapplications of the calculus of probabilities which have made it the real opprobrium of mathematics. [Mill, 1882, Bk. III, ch. XVIII, §3]

The misapplications Mill has in mind are the eighteenth and nineteenth century attempts to compute the credibility of witnesses and the reliability of jury verdicts. His complaint about them is a complaint about averages.

[T]he employment of it for such a purpose implies a misapprehension of the use of averages, which serve, indeed, to protect those whose interest is at stake, against mistaking the general result of large masses of instances, but are of extremely small value as grounds of expectation in any one individual

instance, unless the case be one of those in which the great majority of individual instances do not differ much from the average. [Mill, 1882, Bk. III, ch. XVIII, §3]

That is the distinction between class probability and case probability, stated a century before von Mises [1998] stated it. Mill locates the difficulty where Mises locates it, in the gap between a frequency in a class and a ground for expectation about one member of it.

2.2 Mill changed his mind

The first edition of 1843 attacked the Laplacean theory. Mill withdrew the attack.

This view of the subject was taken in the first edition of the present work; but I have since become convinced that the theory of chances, as conceived by Laplace and by mathematicians generally, has not the fundamental fallacy which I had ascribed to it. [Mill, 1882, Bk. III, ch. XVIII, §1]

Skyrms [2018] traces the change to Mill's correspondence with John Herschel and places it between the first edition and the second edition of 1846, hence before Herschel's review of Quetelet [Herschel, 1850]. What Mill put in place of the attack is an epistemic reading of probability.

We must remember that the probability of an event is not a quality of the event itself, but a mere name for the degree of ground which we, or some one else, have for expecting it. The probability of an event to one person is a different thing from the probability of the same event to another, or to the same person after he has acquired additional evidence.... Every event is in itself certain, not probable; if we knew all, we should either know positively that it will happen, or positively that it will not. But its probability to us means the degree of expectation of its occurrence, which we are warranted in entertaining by our present evidence. [Mill, 1882, Bk. III, ch. XVIII, §1]

Mises quotes the phrase from a chapter whose author had abandoned the position that the phrase is usually taken to support.

2.3 Mill's urn

Mill argues §2 from an urn of unknown composition. A box holds balls of three colors, white, black, and red, mixed in proportions the chooser does not know. Mill concludes that the chooser should lay two to one against white, and he adds a case that sharpens the problem.

[W]e might, on the contrary, know by authentic information that the box contained ninety-nine balls of one color, and only one of the other; still, if we are not told which color has only one, and which has ninety-nine, the drawing of a white and of a black ball will be equally probable to us. [Mill, 1882, Bk. III, ch. XVIII, §2]

Ellsberg [1961] put a three-color urn of unknown composition to the opposite use. Mill treats indifference as settling the matter. Ellsberg used the same apparatus to show that people decline to treat it as settled.

2.4 Mill on statistical laws in the social sciences

Book VI, Chapter IX of the *Logic* states the condition under which a statistical regularity may be carried to a new case.

But those immediate causes depend on remote causes; and the empirical law, obtained by this indirect mode of observation, can only be relied on as applicable to unobserved cases, so long as there is reason to think that no change has taken place in any of the remote causes on which the immediate causes depend. In making use, therefore, of even the best statistical generalizations for the purpose of inferring (though it be only conjecturally) that the same empirical laws will hold in any new case, it is necessary that we be well acquainted with the remoter causes, in order that we may avoid applying the empirical law to cases which differ in any of the circumstances on which the truth of the law ultimately depends. And thus, even where conclusions derived from specific observation are available for practical inferences in new cases, it is necessary that the deductive science should stand sentinel over the whole process. [Mill, 1882, Bk. VI, ch. IX, §5]

The next section denies that enough data can be assembled to average over the combinations that matter.

Now the number of instances necessary to exhaust the whole round of combinations of the various influential circumstances, and thus afford a fair average, never can be obtained. Not only we can never learn with sufficient authenticity the facts of so many instances, but the world itself does not afford them in sufficient numbers, within the limits of the given state of society and civilization which such inquiries always presuppose. [Mill, 1882, Bk. VI, ch. IX, §6]

2.5 Mill's remedy

Mill and Mises identify the same difficulty. A fitted regularity holds only while the causes behind it hold, and the analyst cannot verify that they hold. Their remedies differ. Mill asks for better observations and for a deductive science that stands sentinel over the inference. Mises concludes that quantification in economics is impossible. Mill's remedy is the one that a decision maker who distrusts his model can act on.

3 Constant Relations and Measurement

Chapter II, section 10 of *Human Action* contains the argument on which the rest depends.

There are, in the field of economics, no constant relations, and consequently no measurement is possible. If a statistician determines that a rise of 10 per cent in the supply of potatoes in Atlantis at a definite time was followed by a fall of 8 per cent in the price, he does not establish anything about what happened or may happen with a change in the supply of potatoes in another country or at another time. He has not “measured” the “elasticity of demand” of potatoes. He has established a unique and individual historical fact. [von Mises, 1998, p. 55]

The impracticability of measurement is not due to the lack of technical methods for the establishment of measure. It is due to the absence of constant relations. If it were only caused by technical insufficiency, at least an approximate estimation would be possible in some cases. But the main fact is that there are no constant relations. Economics is not, as ignorant positivists repeat again and again, backward because it is not “quantitative.”

It is not quantitative and does not measure because there are no constants. Statistical figures referring to economic events are historical data. They tell us what happened in a nonrepeatable historical case. [von Mises, 1998, p. 56]

4 Econometrics Named

4.1 The motto of the Econometric Society

Chapter XVI, section 5 of *Human Action* treats three varieties of mathematical economics. The first is the statistical variety.

The first variety is represented by the statisticians who aim at discovering economic laws from the study of economic experience. They aim to transform economics into a “quantitative” science. Their program is condensed in the motto of the Econometric Society: Science is measurement. [von Mises, 1998, p. 347]

Statistics is a method for the presentation of historical facts concerning prices and other relevant data of human action. It is not economics and cannot produce economic theorems and theories. The statistics of prices is economic history.... Nobody ever was or ever will be in a position to observe a change in one of the market data *ceteris paribus*. There is no such thing as quantitative economics. All economic quantities we know about are data of economic history. [von Mises, 1998, pp. 347–348]

Mises then takes up Schultz [1938] on the elasticity of demand and concludes that Schultz’s results “do not refer to potatoes in general, but to potatoes in the United States in the years from 1875 to 1929” [von Mises, 1998, p. 349]. Later in the same chapter he contrasts economics with mechanics.

Our modern industrial civilization is mainly an accomplishment of this utilization of the differential equations of physics. No such constant relations exist, however, between economic elements. The equations formulated by mathematical economics remain a useless piece of mental gymnastics and would remain so even if they were to express much more than they really do. [von Mises, 1998, p. 351]

4.2 Mises quotes the Cowles Commission

The Introduction to *Theory and History* contains the passage closest to our subject. Mises quotes the Cowles Commission's own account of its assumptions.

The econometrician is unable to disprove this fact, which cuts the ground from under his reasoning. He cannot help admitting that there are no "behavior constants." Nonetheless he wants to introduce some numbers, arbitrarily chosen on the basis of a historical fact, as "unknown behavior constants." The sole excuse he advances is that his hypotheses are "saying only that these unknown numbers remain reasonably constant through a period of years." [von Mises, 1957a, p. 11]

The internal quotation is from [Cowles Commission for Research in Economics \[1949\]](#), page 7. Mises continues:

Now whether such a period of supposed constancy of a definite number is still lasting or whether a change in the number has already occurred can only be established later on... It is the assertion of a historical fact, not of a constant that can be resorted to in attempts to predict future events... Nonhuman entities react according to regular patterns; man chooses. [von Mises, 1957a, p. 11]

This paragraph states the parameter-instability half of the argument later made in [Lucas \[1976\]](#). Mises directs it at the Cowles program and sources it to a Cowles document.

4.3 The 1962 formulation

The Ultimate Foundation of Economic Science states the case at its sharpest.

Deluded by the idea that the sciences of human action must ape the technique of the natural sciences, hosts of authors are intent upon a quantification of economics... Their motto is the positivistic maxim: Science is measurement. Supported by rich funds, they are busy reprinting and rearranging statistical data provided by governments, by trade associations, and by corporations and other enterprises. They try to compute the arithmetical relations among various of these data and thus to determine what they call, by analogy with the natural sciences, correlations and functions. They fail to realize that

in the field of human action statistics is always history... As a method of economic analysis econometrics is a childish play with figures that does not contribute anything to the elucidation of the problems of economic reality. [von Mises, 1962, p. 63]

5 Forecasting

Chapter 4 of *The Ultimate Foundation of Economic Science* treats uncertainty directly.

Technological engineering does not eliminate the aleatory element of human existence; it merely restricts its field a little. There always remains an orbit that to the limited knowledge of man appears as an orbit of pure chance and marks life as a gamble. Man and his works are always exposed to the impact of unforeseen and uncontrollable events. [von Mises, 1962, pp. 64–65]

Mises notes that a published forecast changes the process it forecasts.

Forecasting may prove mistaken if it induces men to proceed successfully in a way that is designed to avoid the happening of the forecast events... If people are told in May that the boom going on will continue for several months and will not end in a crash before December, they will try to sell as soon as possible, at any rate before December. Then the boom will come to an end before the day indicated by the prediction. [von Mises, 1962, p. 66]

He grants economics a predictive role and bounds it.

Economics can predict the effects to be expected from resorting to definite measures of economic policies... But, of course, this prediction can be only “qualitative.” It cannot be “quantitative” as there are no constant relations between the factors and effects concerned. [von Mises, 1962, p. 67]

6 Mises and the Statisticians

Richard von Mises (1883–1953), Ludwig’s younger brother, developed the frequency theory of probability from 1919 onward [von Mises, 1957b]. Ludwig accepted that theory for class probability. His claim is not that frequency probability is wrong but

that it says nothing about singular events. The editorial apparatus of the Scholar's Edition argues that Chapter VI of *Human Action* was stimulated by Richard's work [von Mises, 1998].

Gerhard Tintner and Oskar Morgenstern both attended the *Privatseminar* that Mises conducted at the Vienna Chamber of Commerce [Haberler, 1961, Hülsmann, 2007]. Tintner became one of the founders of econometrics as a field [Tintner, 1952]. Several sources describe Mises as the doctoral supervisor of both men. Mises held no salaried chair at Vienna; he was *Privatdozent* from 1913 and unsalaried *ausserordentlicher Professor* from 1918, and other accounts name Hans Mayer as Morgenstern's principal academic supervisor. The seminar attendance is documented. The supervision should be checked against Hülsmann [2007] and the Vienna faculty records before it is asserted in print.

7 What Mises Claimed

Mises made three claims that should be kept separate.

First, economic time series record singular historical events rather than repeated draws from a stable law. A fitted coefficient summarises a time and a place. Mises is right about this, and the point is not a minor one. It is the same observation that motivates Lucas [1976] and that motivates our treatment of a decision maker who does not know the data generating process [Hansen and Sargent, 2008].

Second, the analyst who writes down a constant is making an assumption about the future that his data cannot support. The Cowles passage in *Theory and History* is the strongest form of the argument, because there Mises is not asserting an impossibility. He is pointing at a specific assumption that the practitioners themselves wrote down.

Third, quantification in economics is therefore impossible. This claim is stronger than the first two and does not follow from them. A decision maker who distrusts his model need not abandon numbers. He can carry a set of models and act well against the worst of them. That is the program in Hansen and Sargent [2008], and it addresses the problem that Mises identified rather than the conclusion he drew from it.

Mises worked before the tools for the third step existed. Frank Knight's distinction between measurable risk and unmeasurable uncertainty [Knight, 1921] was the state of

the art when Mises wrote Chapter VI, and Mises sharpened it by locating the difficulty in the singularity of the event rather than in the analyst's ignorance. He posed the problem well. We should say so, and we should say that the answer we give is one he did not have available.

Mill reached the same difficulty and drew the opposite conclusion. Quoting Mill on the opprobrium of mathematics alongside Mises requires reporting that Mill withdrew his attack on the Laplacean theory, and that Mill asked for better data under the supervision of theory rather than for less measurement [Mill, 1882, Bk. III, ch. XVIII; Bk. VI, ch. IX].

References

- Cowles Commission for Research in Economics. Report for period, january 1, 1948–june 30, 1949. Technical report, University of Chicago, Chicago, 1949. Quoted at [von Mises \[1957a\]](#), p. 11].
- Daniel Ellsberg. Risk, ambiguity, and the savage axioms. *Quarterly Journal of Economics*, 75(4):643–669, 1961.
- Gottfried Haberler. Mises's private seminar: Reminiscences. *The Mont Pelerin Quarterly*, 1961. Reprinted by the Ludwig von Mises Institute.
- Lars Peter Hansen and Thomas J. Sargent. *Robustness*. Princeton University Press, Princeton, 2008.
- John F. W. Herschel. Quetelet on probabilities. *Edinburgh Review*, 92:1–57, 1850.
- Jörg Guido Hülsmann. *Mises: The Last Knight of Liberalism*. Ludwig von Mises Institute, Auburn, Alabama, 2007.
- Frank H. Knight. *Risk, Uncertainty and Profit*. Houghton Mifflin, Boston, 1921.
- Jr. Lucas, Robert E. Econometric policy evaluation: A critique. In Karl Brunner and Allan H. Meltzer, editors, *The Phillips Curve and Labor Markets*, volume 1 of *Carnegie-Rochester Conference Series on Public Policy*, pages 19–46. North-Holland, Amsterdam, 1976.

John Stuart Mill. *A System of Logic, Ratiocinative and Inductive: Being a Connected View of the Principles of Evidence and the Methods of Scientific Investigation*. Harper and Brothers, New York, 8th edition, 1882. First edition 1843. Mises cites a 1936 London impression, p. 353. References here are to book, chapter, and section, which are stable across editions.

Henry Schultz. *The Theory and Measurement of Demand*. University of Chicago Press, Chicago, 1938.

Brian Skyrms. Mill's conversion: The herschel connection. *Philosophers' Imprint*, 18 (23):1–24, 2018.

Gerhard Tintner. *Econometrics*. John Wiley and Sons, New York, 1952.

Ludwig von Mises. *Theory and History: An Interpretation of Social and Economic Evolution*. Yale University Press, New Haven, 1957a.

Ludwig von Mises. *The Ultimate Foundation of Economic Science: An Essay on Method*. D. Van Nostrand, Princeton, 1962.

Ludwig von Mises. *Human Action: A Treatise on Economics*. Ludwig von Mises Institute, Auburn, Alabama, scholar's edition, 1998. Reprint of the first edition, Yale University Press, New Haven, 1949. Page references in this note are to the Scholar's Edition.

Richard von Mises. *Probability, Statistics and Truth*. Dover, New York, 2nd revised english edition, 1957b. First German edition, *Wahrscheinlichkeit, Statistik und Wahrheit*, Julius Springer, Vienna, 1928.